

AnalystLab Africa Data Analytics Internship Program

Week 3: SQL & Data Querying

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Task 1: Chinook Database

Objective

The objective of this task was to analyze customer purchasing behavior, revenue trends, and product performance using SQL queries.

Query Categories

Basic Queries

SQL concepts used:

- SELECT
- WHERE
- ORDER BY

Aggregate Functions

SQL concepts used:

- COUNT()
- SUM()
- AVG()
- GROUP BY
- HAVING

Joins

Tables were joined to analyze customer purchases, track sales, and artist performance.

Subqueries

Subqueries were used to identify maximum values and above-average performance.

Window Functions

RANK(), ROW_NUMBER(), and PARTITION BY were used to rank customers and invoices.

Business Insights

Top Customers

The analysis identified the highest-spending customers, which are valuable for customer retention strategies.

Revenue Trends

Revenue analysis revealed changes in sales performance over time.

Top Selling Tracks

The most popular tracks and artists were identified, providing insights into customer preferences.

Customer Purchasing Behavior

Customer spending patterns and average purchase values were analyzed to understand purchasing habits.

Task 2: Sales Dataset

Objective

The objective of this task was to analyze sales performance, customer behavior, and product trends using SQL.

Query Categories

Basic Queries

SELECT, WHERE, and ORDER BY were used to explore the data.

Aggregate Functions

SUM(), AVG(), COUNT(), GROUP BY, and HAVING were used to summarize sales performance.

Subqueries

Subqueries were used to identify the highest sales and above-average transactions.

Window Functions

RANK(), ROW_NUMBER(), and PARTITION BY were used to rank customers and orders.

Business Insights

Top Customers

The highest-value customers were identified based on total spending.

Product Performance

Product lines generating the highest revenue were identified.

Revenue Trends

Sales trends across years were analyzed to evaluate growth patterns.

Geographic Analysis

Countries contributing the most revenue were identified.

Deal Size Analysis

The contribution of different deal sizes to total revenue was evaluated.

Customer Purchasing Behavior

Customer order frequency and average order value were analyzed to understand purchasing patterns.